

Introduction to BASIC Programming Language:

Meaning of BASIC:

BASIC stands for **B**eginner's **A**ll-purpose **S**ymbolic **I**nstruction **C**ode. It was developed in the 1960s at Dartmouth College by John Kemeny and Thomas Kurtz. BASIC was designed to be easy for beginners to learn and use, making it popular for educational purposes and early personal computing.

BASIC Character Set

The BASIC character set includes:

Alphabetic (Letters: A-Z)

Numeric (Digits: 0-9)

Special Characters: Symbols like `+`, `-`, `\*`, `/`, `=`, `<`, `>`, `(`, `)`, `,`, `;`, `.` , `!`, `"` , `:`, `?`, `!`, `#`, `\$`, `%`, `&`, `\_`.

Keywords: Reserved words like `PRINT`, `LET`, `INPUT`, `IF`, `THEN`, `END`, `READ`, `DATA`.

BASIC Statements

READ and DATA: The `READ` statement is used to read values from `DATA` statements. `DATA` statements provide the data that `READ` will use.

Example:

```
` ``basic
10 READ A, B
20 DATA 10, 25
` ``
```

LET: The `LET` statement is used to assign a value to a variable.

Example:

```
` ``basic
30 LET X = 100
` ``
```

(The `LET` keyword is often optional in many BASIC dialects).

INPUT: The `INPUT` statement allows the user to enter data while the program is running.

Example:

```
` `` `basic
40 INPUT "Enter your age: ", AGE
` `` `
```

BASIC Arithmetic Operations/Expressions

BASIC uses standard arithmetic operators:

`+` : Addition

`-` : Subtraction

`\*` : Multiplication

`/` : Division

`^` : Exponentiation (e.g., `X^2` for X squared)

Expressions combine variables, constants, and operators.

Example: `Y = (A + B) \* 5`

Simple BASIC Program

Here's a very simple program to add two numbers entered by the user:

```
` `` `basic
10 REM A simple addition program
20 INPUT "Enter the first number: ", NUM1
30 INPUT "Enter the second number: ", NUM2
40 SUM = NUM1 + NUM2
50 PRINT "The sum is: "; SUM
60 END
` `` `
```

Line numbers (10, 20, etc.) indicate the order of execution.

`REM` is for remarks or comments, ignored by the computer.

The program prompts the user for two numbers, adds them, and displays the result.